

lecture - 35: calculating the classification loss and accuracy

6.6 Calculating the classification loss and accuracy

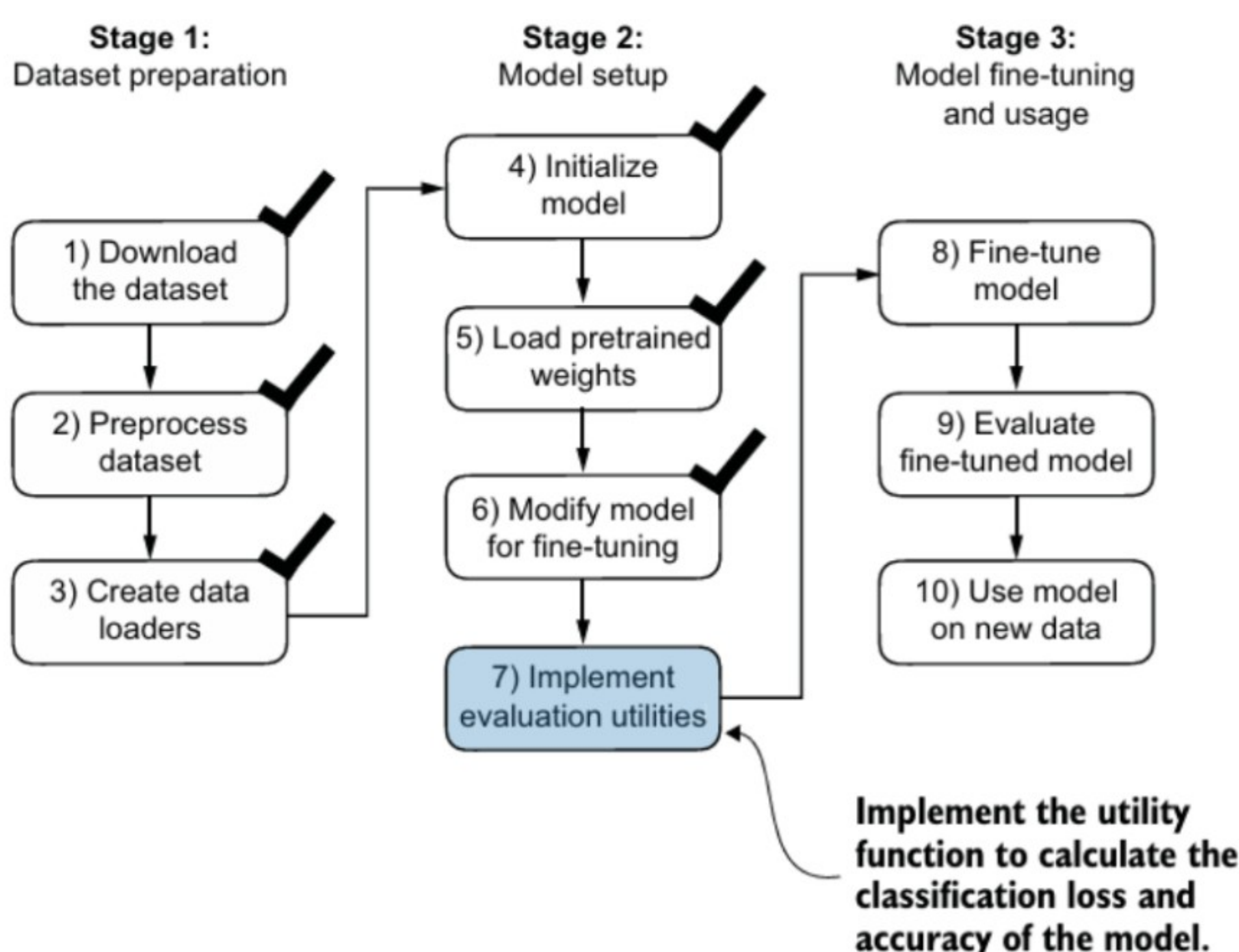


Figure 6.13 The three-stage process for classification fine-tuning the LLM. We've completed the first six steps. We are now ready to undertake the last step of stage 2: implementing the functions to evaluate the model's performance to classify spam messages before, during, and after the fine-tuning.

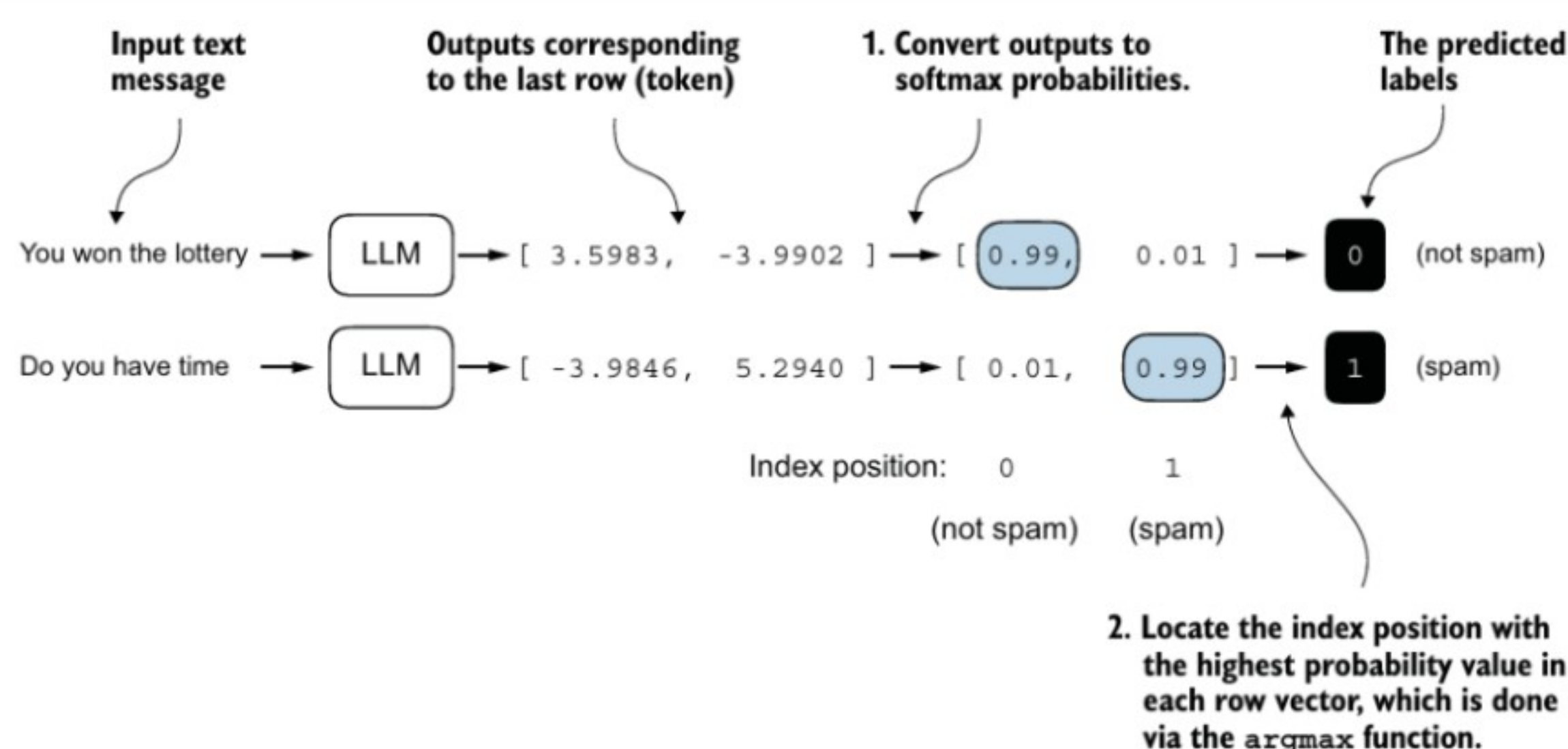
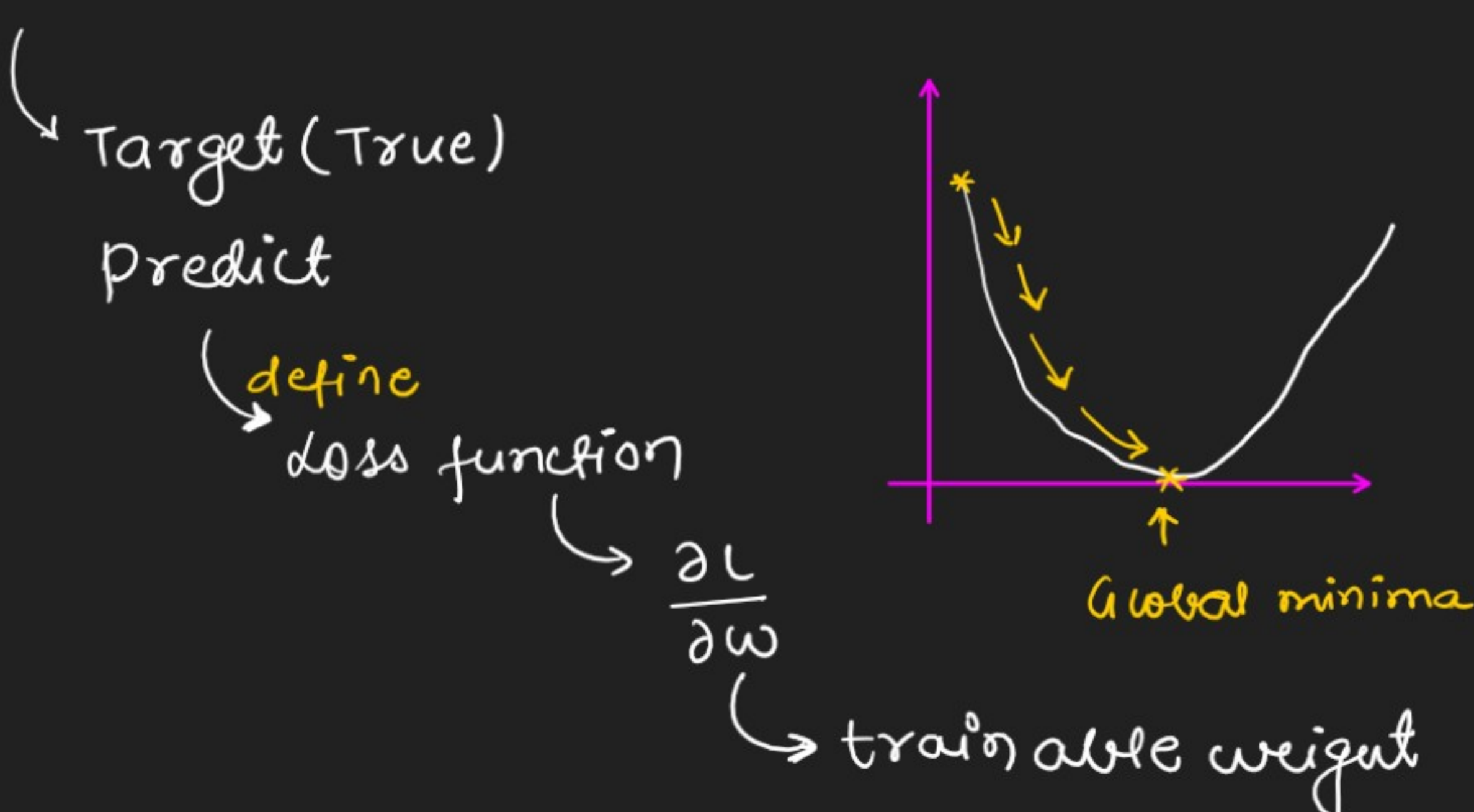


Figure 6.14 The model outputs corresponding to the last token are converted into probability scores for each input text. The class labels are obtained by looking up the index position of the highest probability score. The model predicts the spam labels incorrectly because it has not yet been trained.



$$W_{\text{new}} = W_{\text{old}} - \frac{\partial L}{\partial W_{\text{old}}}$$

From neural network

Prediction: p_i

Target: y_i

so, categorical entropy loss:

$$-\sum y_i \log p_i$$

True value

Predict value

loss function: cross entropy loss

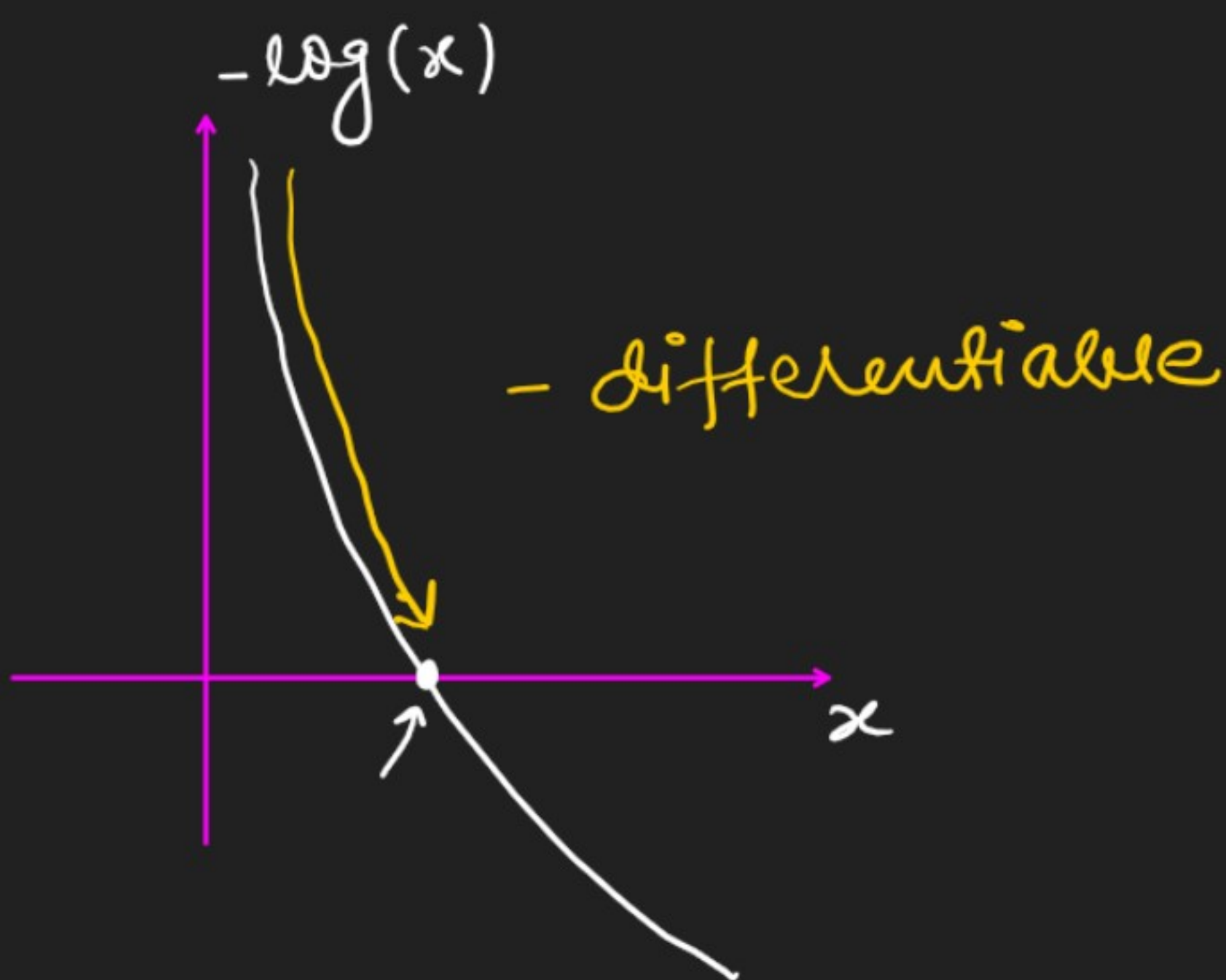
True: non-spam ([1, 0])

Predict: [0.8, 0.2]
 ↓ ↘
 non-spam spam

$$\text{cross entropy loss} = - \sum_{i=1}^n y_i \log(p_i)$$

$$\Rightarrow - [1 * \log(0.8) + 0 * \log(0.2)]$$

$$= 0.2231$$



Step 6: Finetune model on supervised data

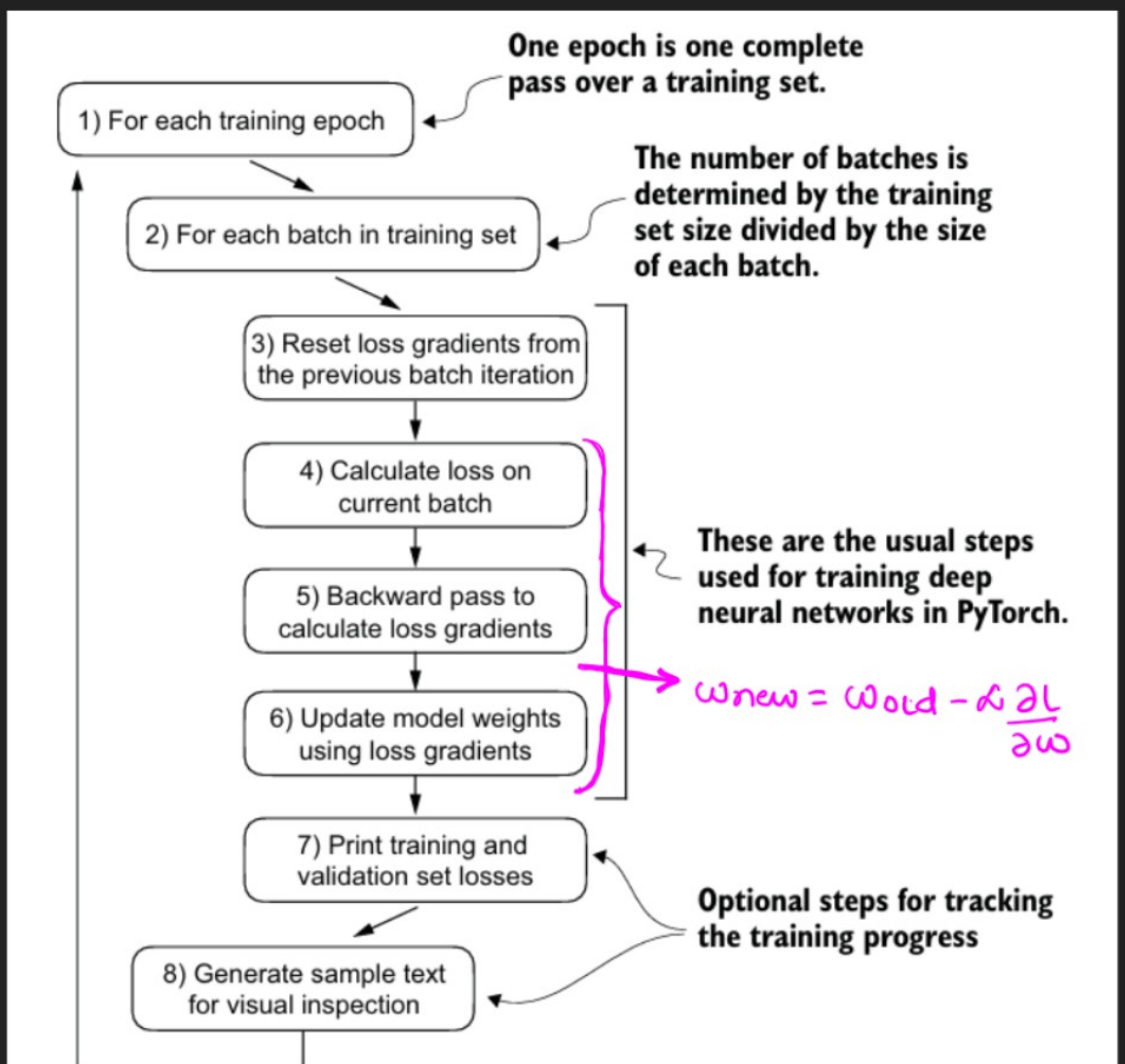


Figure 6.15 A typical training loop for training deep neural networks in PyTorch consists of several steps, iterating over the batches in the training set for several epochs. In each loop, we calculate the loss for each training set batch to determine loss gradients, which we use to update the model weights to minimize the training set loss.